

WT Module 4

Give the significance of WEP protocols. What are the features of WPA2?

Significance of WEP Protocols

WEP (Wired Equivalent Privacy) is a security protocol used in wireless LANs to provide privacy and protect data transmission. It was designed to give wireless networks security similar to wired networks.

Key Features & Significance

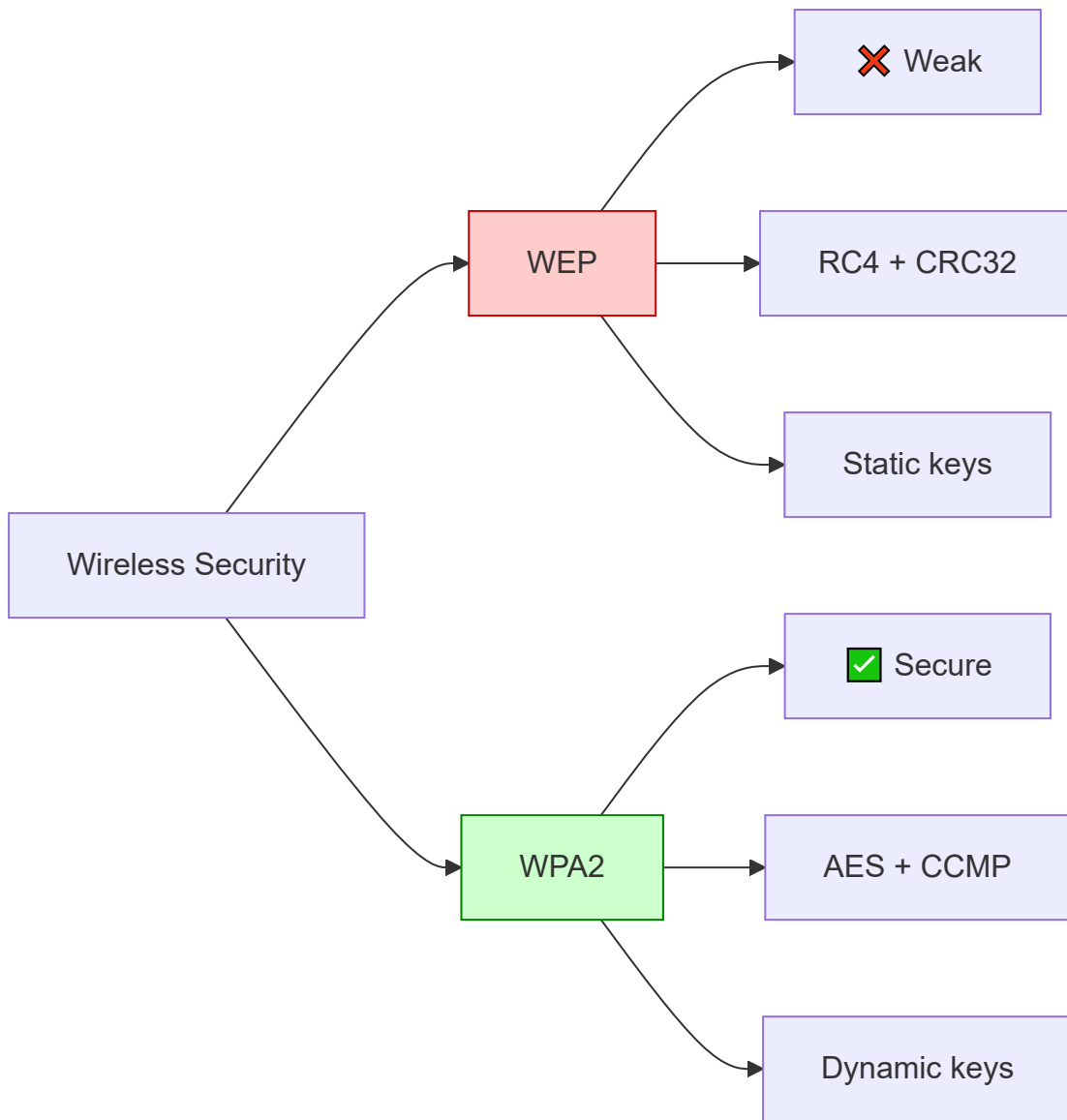
- **Data Encryption:** WEP encrypts data before transmission to prevent unauthorized access.
- **Basic encryption** using RC4 stream cipher
- **40-bit or 104-bit keys** with a 24-bit Initialization Vector (IV), Total uses 64-bit and 128-bit keys.
- **Shared-key authentication** for access control
- **Basic data integrity** using CRC-32 checksum
- Helped establish the **foundation for wireless security standards**, though later found weak and easily cracked

Features of WPA2

WPA2 (Wi-Fi Protected Access 2) is the successor to WPA, introduced by IEEE to overcome WEP's weaknesses. It is considered the most secure standard until WPA3.

Key Features

- **Strong Encryption:** Uses AES (Advanced Encryption Standard) with CCMP for confidentiality and integrity
- **Robust Authentication:** Supports IEEE 802.1X with EAP for enterprise networks
- **Data Integrity:** Prevents packet tampering with strong message integrity checks
- **Dynamic Key Management:** Keys are generated and distributed securely, reducing risk of reuse
- **High Security Standard:** Provides strong protection for both personal and enterprise WLANs



Security Standards Used After WEP

1. WPA (Wi-Fi Protected Access)

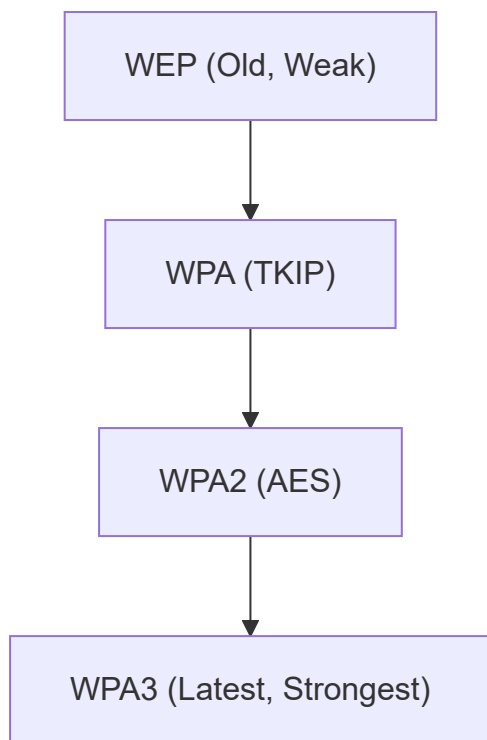
- Improved security over WEP
- Uses **TKIP** (Temporal Key Integrity Protocol) for better encryption

2. WPA2

- Stronger security standard
- Uses **AES** (Advanced Encryption Standard) encryption

3. WPA3

- Latest and more secure standard
- Provides stronger authentication and protection



Discuss the flaws in WEP's initialization vector (IV) implementation and how it leads to key reuse and susceptibility to attacks.

- **WEP (Wired Equivalent Privacy)** was the first security protocol for IEEE 802.11 WLANs
- Uses the **RC4 stream cipher** with a secret key and a 24-bit **Initialization Vector (IV)**
- The IV is meant to ensure that each packet is encrypted uniquely

Flaws in IV Implementation

1. Small IV Size (24-bit)

- Only $2^{24} \approx 16$ million possible IVs
- In busy networks, IVs repeat quickly (within hours)

2. IV Sent in Cleartext

- IV is transmitted openly in each packet header
- Attackers can easily capture IVs without breaking encryption

3. Weak RC4 Key Scheduling

- WEP combines IV with the secret key to form the RC4 seed
- Certain IV values produce weak keys, making it easier to recover the secret key

4. Key Reuse

- Because IVs repeat, the same RC4 keystream is reused
- Attackers can compare ciphertexts encrypted with the same keystream to deduce plaintext

Consequences / Susceptibility to Attacks

- **Statistical Analysis Attacks:** Repeated IVs allow attackers to analyze patterns and recover the key
- **Known Plaintext Attacks:** If part of the plaintext is known (e.g., headers), attackers can exploit IV reuse to recover keystreams
- **Key Recovery Attacks (e.g., FMS Attack):** Exploits weak IVs to recover the WEP key in minutes
- **Practical Breakability:** Tools like Aircrack-ng can crack WEP keys by collecting enough packets with reused IVs